Anirudh Ravipudi

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Professional Summary

Data Engineer with three years of experience turning fragmented, multi-cloud data into reliable pipelines that power analytics and ML systems. Built and optimized production workflows on Databricks, Snowflake, Kafka, and Airflow, cutting data preparation time by 30% and accelerating model training readiness by 20% through targeted PySpark and dbt optimization. Focused on data quality, pipeline reliability, and building infrastructure that data science teams can actually trust.

Experience

AI Cloud Solutions — **Aug 2024 - Present**

Data Engineer

- Optimized distributed PySpark jobs using partitioning, Z-ordering, and parallel processing strategies, reducing data preparation time by 30% and cutting downstream job runtime across core pipelines.
- Partnered with Data Science teams to build behavioral feature stores with rolling aggregations and anomaly indicators, cutting downstream model training preparation time by 20%.
- Designed and implemented cross-cloud data ingestion pipelines using Kafka and batch AWS S3 loads, streaming transactional datasets from GCP and Azure sources into a core AWS Delta Lakehouse with low-latency availability for downstream microservices.
- Built automated data quality checks integrated into Apache Airflow pipelines, catching upstream schema drift and anomalies before they reached production tables and reducing manual data validation effort.
- Designed a modular transformation layer using dbt and Snowflake, building reusable dimension and fact tables (Kimball modeling) to standardize reporting logic across multiple downstream workflows.
- Containerized pipeline microservices using Docker and Kubernetes, and set up automated CI/CD deployment workflows via Jenkins and GitHub, shortening deployment turnaround.
- Contributed to platform governance by implementing access control and data lineage tracking in Databricks Unity Catalog, and tuned Snowflake warehouse sizing to help reduce compute costs.

FoxFire Technologies — **Jan 2021 - Dec 2021**

Data Analyst Intern

- Optimized data extraction scripts and streamlined daily ETL processes from relational databases, reducing report generation latency by 15%.
- Built enterprise operational dashboards in Tableau and Power BI, centralizing supply chain and inventory metrics for real-time stakeholder visibility.
- Developed an automated data quality framework using SQL and Python, using statistical anomaly detection to isolate corrupted ingestion records before they reached production analytics.
- Conducted root-cause analyses on pipeline discrepancies using hypothesis testing to identify edge cases in transaction streams, improving downstream data reliability for executive KPI tracking.

Projects

Fraud Detection Data Pipeline & ML System

- Trained Random Forest and XGBoost classifiers to detect fraud, achieving 0.98 ROC-AUC while handling severe class imbalance (0.17% fraud rate); selected Random Forest at 83% recall and 77% precision on 56,962 held-out transactions.
- Deployed the fraud prediction model via a FastAPI REST endpoint, containerized with Docker and managed on Kubernetes, returning fraud labels and probability scores within 200ms.
- Built a scalable PySpark ETL pipeline orchestrated with Airflow to ingest and transform 284,807 real credit card transactions from PostgreSQL, writing processed Parquet files to AWS S3 for downstream ML workflows.
- Engineered 5 behavioral features, log-normalized transaction amounts, rolling transaction velocity, rolling average amount, transaction hour, and a high-value transaction flag, to improve model signal quality.

LLM Financial Intelligence Platform

- Built data pipelines ingesting and processing 28,877 stock price records and 454 quarterly earnings records across 10 years of history, structured for downstream analysis and modeling.
- Trained a GradientBoosting model on historical pricing and earnings data to generate directional predictions, and built a visualization pipeline producing 14 interactive Plotly charts.
- Implemented abstractive summarization using facebook/bart-large-cnn and financial sentiment analysis using ProsusAI/finbert on earnings call transcripts, correctly identifying Apple as the only negative report at 94.88% confidence.
- Exposed the full pipeline via a FastAPI REST API with 9 endpoints covering price data, earnings history, predictions, and sentiment analysis, deployed on Render.

A/B Testing & Experimentation Pipeline

- Built an end-to-end experimentation platform in Python simulating an e-commerce checkout A/B test across 10,000 synthetic users over a 14-day period.
- Applied z-tests for statistical significance and computed confidence intervals and lift metrics, measuring a 26% conversion lift ($p = 0.000007$).
- Exposed simulation and results via a FastAPI REST API backed by PostgreSQL, supporting on-demand test runs.

Technical Skills

- **Programming & Scripting**: Python, SQL, R, Java, C/C++
- **Data Engineering & Streaming**: Apache Spark, PySpark, Airflow, Kafka, ELT/ETL Pipelines, Data Modeling (Kimball/Dimensional), Delta Lake, Ingestion Frameworks
- **Cloud & Infrastructure**: AWS (S3, EC2, IAM, Redshift), Kubernetes, Docker, Azure & GCP (Cross-Cloud Integration/Data Egress)
- **Data Warehousing & Transformations**: Snowflake, Databricks (Unity Catalog), dbt, PostgreSQL, Redshift
- **Visualization & Tools**: Power BI, Tableau, Looker, Git, GitHub, Jenkins, CI/CD
- **Analytics & Machine Learning**: Feature Engineering, NumPy, pandas, scikit-learn, Statistical Methods, FastAPI

Education

Florida International University — **2022 - 2024**

Master's, Data Science & Artificial Intelligence — *Miami, FL*